

Hernán Salinas Castro

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Interests

PhD student in Astronomy with experience in exoplanet characterization through the transit method and Markov Chain Monte Carlo (MCMC) techniques. Currently focused on the study of the chemical composition of protoplanetary disks using observations from the James Webb Space Telescope (JWST). My goal is to advance our understanding of the chemistry of exoplanetary atmospheres and protoplanetary environments to better constrain planet formation processes.

Education

Pontificia Universidad Católica de Chile <i>PhD in astrophysics</i>	Santiago, Chile 2026 - Present
Universidad Andrés Bello <i>Bachelor's degree in astronomy</i>	Santiago, Chile 2020 - 2024

Research experience

Chemistry of FU Orionis Objects with JWST	March 2026-Present
Investigation of FU Orionis (FUor) disks, focusing on the chemical composition of inner disk regions following accretion outbursts using JWST data under the supervision Dr. Viviana Guzmán.	

Visitor at ESO (Santiago)	June 2025
Study of exoplanetary atmospheres through high-resolution spectroscopy techniques. Conducted calibration and data analysis of observations obtained with the HARPS and NIRPS spectrographs, focusing on a hot Jupiter-type exoplanet in a close-in orbit around its host star. This work is carried out under the supervision of Dr. Elyar Sedaghati.	

VLTI Observations of the Atmosphere of Normal Red Giants	January 2025 - June 2025
Used AMBER data from the VLTI to study red giant stars through optical and infrared interferometry techniques. During this project, I had the opportunity to learn how to calibrate and analyze interferometric data under the supervision of Dr. Keiichi Ohnaka.	

Member of the MOTE Project Monitoring of Transiting Exoplanets (MOTE) Project	December 2023 – Present
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Participated in a project focused on training undergraduate students in the analysis of exoplanet transit data. I contributed to the selection and planning of observation nights for target exoplanets using different telescopes, such as SPECULOOS and LCOGT. I was also involved in the calibration and reduction of the collected data, followed by photometric analysis. As part of this work, I developed a sub-project focused on the study of Jupiter-type exoplanets orbiting M-dwarf stars, which became the topic of my undergraduate thesis. Additionally, I had the opportunity to write an observation proposal during my undergraduate studies as part of this sub-project. This entire project was carried out under the supervision of Dr. Claudio Cáceres.

Workshops and courses

ALMA Cycle 13 Proposal Preparation <i>Silver Jubilee: Exoplanet Demographics</i>	Valparaiso, Chile March 20, 2026
NASA Exoplanet Science Institute <i>Silver Jubilee: Exoplanet Demographics</i>	Virtual July 21-25, 2025
European Interferometry Initiative (EII) <i>Horizons for Optical Long Baseline Interferometry</i>	Virtual January 27-30, 2025
Universidad Andrés Bello <i>Radio astronomy course</i>	Santiago, Chile August - December, 2024
Universidad Andrés Bello <i>Complementary cosmological simulations course</i>	Santiago, Chile August - December, 2024
Space Mood <i>Space Biology Networking in Peru (BEP)</i>	Virtual December 14, 2024
Universidad Andrés Bello <i>First Astrochemistry Workshop</i>	Santiago, Chile October 28, 2024
NASA Exoplanet Science Institute <i>Advances in Direct Imaging: From Young Jupiters to Habitable Earths</i>	Virtual July 22-26, 2024
24RED <i>The virtual astrobiology introductory course</i>	Virtual June 2024
Centro de Astrofísica y Tecnologías Afines (CATA) <i>CATA Area 6 (Exoplanetas y astrobiología) Meeting</i>	Santiago, Chile May 7, 2024
Universidad Andrés Bello <i>Complementary interferometry course</i>	Santiago, Chile March - June 2024

Talks

Centro de Astrofísica y Tecnologías Afines (CATA)

CATA Area 6 Meeting

"Jupiter type exoplanets orbiting M dwarf stars"

Santiago, Chile

August 6, 2025

Outreach activities

Member of the Astroamigos Wintata Wara Group

April 2025 - Present

A astronomy group engaged in public science outreach through activities such as telescope observations, public talks, interactive stations, and workshops. Within the group, I serve as a science communicator and collaborator, supporting content development and assisting with telescope operations. I have participated in events at institutions such as Observatorio Calán, Observatorio Manuel Foster, and Museo Nacional Aeronáutico y del Espacio. Instagram: @astroamigos_wintata_wara.

Galaxies for Children Workshop

May 2023

Together with two colleagues, I organized an outreach workshop for children at Luis Cruz Martínez School in Andacollo, Chile. We introduced basic concepts about galaxies and guided an interactive activity where the children created their own galaxies using plasticine. The goal was to foster creativity, curiosity, and interest in astronomy.

Technical Skills

Languages: Spanish - Intermediate English.

Programming language: Python and LaTeX(intermediate level).

Bash script (Basic level).

Software:

Intermediate level: Topcat, Aladin, SAOImageDS9.

Basic level: Excel, Iraf, Astroimagej, Casa, Carta, FV.

TESS telescope data analysis(Basic level).

Photometric data analysis for exoplanets:

Intermediate level: exotic_ld, ccdproc, batman, lightkurve, emcee.